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Circular RNA expression profiles in hippocampus from mice with perinatal glyphosate exposure

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ABSTRACT

Glyphosate is the active ingredient in numerous herbicide formulations. The roles of glyphosate in embryo-toxicity and neurotoxicity have been reported in human and animal models. Recently, several studies have reported evidence linking neurodevelopmental disorders (NDDs) with gestational glyphosate exposure. However, the role of glyphosate in neuronal development is still not fully understood. Our previous study found that perinatal glyphosate exposure resulted in differential microRNA expression in the prefrontal cortex of mouse offspring. However, the mechanism of glyphosate-induced neurotoxicity in the developing brain is still not fully understood. Considering the pivotal role of Circular RNAs (circRNAs) in the regulation of gene expression, a circRNA microarray method was used in this study to investigate circRNA expression changes in the hippocampus of mice with perinatal glyphosate exposure. The circRNA microarrays revealed that 663 circRNAs were significantly altered in the perinatal glyphosate exposure group compared with the control group. Among them, 330 were significantly upregulated, and the other 333 were downregulated. Furthermore, the relative expression levels of mmu-circRNA-014015, mmu-circRNA-28128 and mmu-circRNA-29837 were verified using quantitative real-time polymerase chain reaction (qRT-PCR). Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analyses demonstrated that stress-associated steroid metabolism pathways, such as aldosterone synthesis and secretion pathways, may be involved in the neurotoxicity of glyphosate. These results showed that circRNAs are aberrantly expressed in the hippocampus of mice with perinatal glyphosate exposure and play potential roles in glyphosate-induced neurotoxicity.

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1. Introduction

Glyphosate, a phosphonomethyl amino acid derivative, is commonly used as a broad-spectrum, non-selective, post-emergence herbicide and is widely used in both agricultural and non-agricultural activities, such as gardening [1]. Glyphosate-based herbicides (GBH) are the most utilized agrochemicals in the world [2]. Approximately two-thirds of the total volume of GBH used has been delivered to the environment [3], and it induces soil erosion

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[4]. It has been thought that the use of GBH does not cause adverse effects to mammals [5]. However, recently attention has been drawn to the overuse of glyphosate and its harm to human health. Animal studies suggest that glyphosate may be neurotoxic to the developing brain [6,7]. In vitro studies have revealed that glyphosate can pass through the blood brain barrier and placental barrier [8]. Results from the Childhood Autism Risks from Genetics and Environment study provide further evidence for an association between neurodevelopmental disorders (NDDs) and gestational organophosphate exposure, particularly glyphosate [9]. However, the mechanism of glyphosate-induced neurotoxicity in the developing brain is still not fully understood.

Novel studies have provided insight into the possible roles of environmental and epigenetic factors in NDDs [10,11]. Our previous study showed that alterations of the miRNAs and the target genes associated with neurogenesis and neuron differentiation may play

a role in the neurotoxicity of glyphosate [12]. However, it is likely that other factors are also involved, and these mechanisms are yet to be fully elucidated.

Circular RNAs (circRNAs) are a special class of non-coding RNAs (ncRNAs) which may interact with miRNAs [13]. CircRNAs may act as miRNA sponges and could play an important role in regulating gene expression through a circRNA-miRNA gene pathway. There is increasing evidence that circRNA-mediated gene expression regulation is widespread and critical in mammals. Recent studies reveal a striking regulation of circRNAs during neuronal development [14]. Altered expression of circRNAs may contribute to the post-transcriptional modulation of genes involved in central nervous system diseases [15].

Little is known about the functions of circRNAs in developing brain with perinatal glyphosate exposure. Specifically, there are no reports on the expression profiles of circRNAs in developing brain with perinatal glyphosate exposure. Herein, we established a perinatal glyphosate exposure model and performed microarray detection to profile the expression of circRNAs in the hippocampus from exposed mice. Then, we investigated circRNAs with differential expression after perinatal glyphosate exposure.

2. Materials and methods

2.1. Reagents

The pesticide used in this study was a commercial formulation marketed in China as Roundup® (Monsanto Company, St. Louis, MO, USA), containing 48 g glyphosate isopropylamine salt per 100 cm³ of product (equivalent to 35.6% w/v of glyphosate acid).

2.2. Animals

Qualified specific pathogen-free pregnant ICR mice (aged 9–11 weeks), weighing 40–50 g, obtained from Shanghai Laboratory Animal Center at the Chinese Academy of Sciences (Shanghai, China). The mice were kept for 1 week under controlled temperature and humidity with a day/night cycle and access to standard laboratory food and water *ad libitum*. The study was approved by the Institutional Animal Care and Use Committee of Hangzhou Medical College (Hangzhou, China) and based on the NIH Guide for the Care and Use of Laboratory Animals.

2.3. Perinatal glyphosate exposure model

Sixteen pregnant ICR mice were randomly divided into two groups of eight mice. In the control group, pregnant mice were provided with purified water. In the glyphosate-treated group, pregnant mice were provided with drinking water containing 0.38% glyphosate (1% Roundup®) from embryonic day (E) 14 to postnatal day (PND) 7 (the day of birth considered as PND 0), then provided with normal drinking water. The dose administered was equivalent to 50 mg of glyphosate/kg/day, which was 1/20th of the glyphosate no-observed-adverse-effect level for glyphosate described previously [5]. The offspring received glyphosate indirectly via pregnancy and lactation. Weaning occurred on PND 21, then the offspring were provided with normal drinking water. A total of 8 male offspring from each group were sacrificed on PND 28, the brains were quickly removed, and the hippocampi were isolated on an ice pad.

2.4. RNA isolation

Total RNA was isolated using TRIzol reagent (Invitrogen, Carlsbad, USA) according to the manufacturer's protocol. The

concentrations of the RNA samples were determined by the OD260/OD280 ratio using a NanoDrop ND-1000 instrument (Agilent, Santa Clara, CA, USA). The integrity of the RNA was assessed by electrophoresis on a denaturing agarose gel.

2.5. CircRNA microarray analysis

The sample preparation and microarray hybridization were performed according to Arraystar's standard protocols. The total isolated RNA was taken from pooled samples of each group. Briefly, total RNA was digested with Rnase R (Epicentre, Inc.) to remove linear RNAs and enrich for circular RNA. Then, the enriched circular RNA was amplified and transcribed into fluorescent cRNA utilizing a random priming method (Arraystar Super RNA Labeling Kit; Arraystar). The labeled cRNA was hybridized onto the Arraystar Mouse circRNA Array v2 (8 × 15 K, Arraystar), and the arrays were scanned by the Agilent Scanner G2505C. Agilent Feature Extraction software (version 11.0.1.1, USA) was used to analyze acquired array images. Quantile normalization and subsequent data processing were performed using the limma package in R software. Differentially expressed circRNAs between the two groups were identified through Fold Change filtering. Hierarchical Clustering was performed to show the differential circRNAs expression pattern among samples. circRNAs with an absolute value of fold change ≥ 2.0 and a p-value ≤ 0.05 were selected as significantly differentially expressed.

2.6. Quantitative real-time polymerase chain reaction (qRT-PCR) analysis

After determining the optimal annealing temperatures, qRT-PCR was performed in the ABI PRISM® 7500 Sequence Detection System (Applied Biosystem, Foster City, CA, USA) using SYBR Green qPCR SuperMix (Invitrogen, Carlsbad, CA, USA). The primers used for the qRT-PCR analysis were as follows: 5'-GAGAAGGCTCCGAAAGAATA-3' (sense) and 5'-CACITGGAAGGCTGTAACAT-3' (anti-sense) for mmu-circRNA-014015; 5'-AGGAATTGCCATTGTCAGC-3' (sense), 5'-CCCAGACCATCCACATCAC-3' (anti-sense) for mmu-circRNA-28128; 5'-AACGGGAAAATCCAAAGCAC-3' (sense), 5'-CAGGA-CAGGAAGACAGAACTCG-3' (anti-sense) for mmu-circRNA-29837; and 5'-CACTGAGCAAGAGAGGCCCTAT-3' (sense), 5'-GACG-GAACTTTATTGATGGTATT-3' (anti-sense) for GAPDH. Both the perinatal glyphosate exposure group and the control group had three independent samples and all samples were conducted in triplicate. The PCR conditions were a 95 °C denaturation for 10 min, followed by 40 cycles at 95 °C for 10 s and 60 °C for 60 s. Melt curve analysis was carried out after the PCR to confirm primer specificity. The relative expression of circRNAs was calculated using the 2^{-ΔΔCt} method [16].

2.7. Computational bioinformatics analysis

CircRNA/miRNA interactions were predicted with Arraystar's homemade miRNA target prediction software based on TargetScan (http://www.targetscan.org/vert_71/) and miRanda (www.microrna.org/). Then, a circRNA-miRNA gene network was generated to visualize the interactions using Cytoscape. GO (Gene Ontology) annotation and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses were used to annotate the functions of the targets genes.

2.8. Statistical analysis

All data are presented as the mean $\pm$ standard deviation (SD). Statistical significance was tested by Student's *t*-test, and a P-value < 0.05 was considered to be statistically significant.

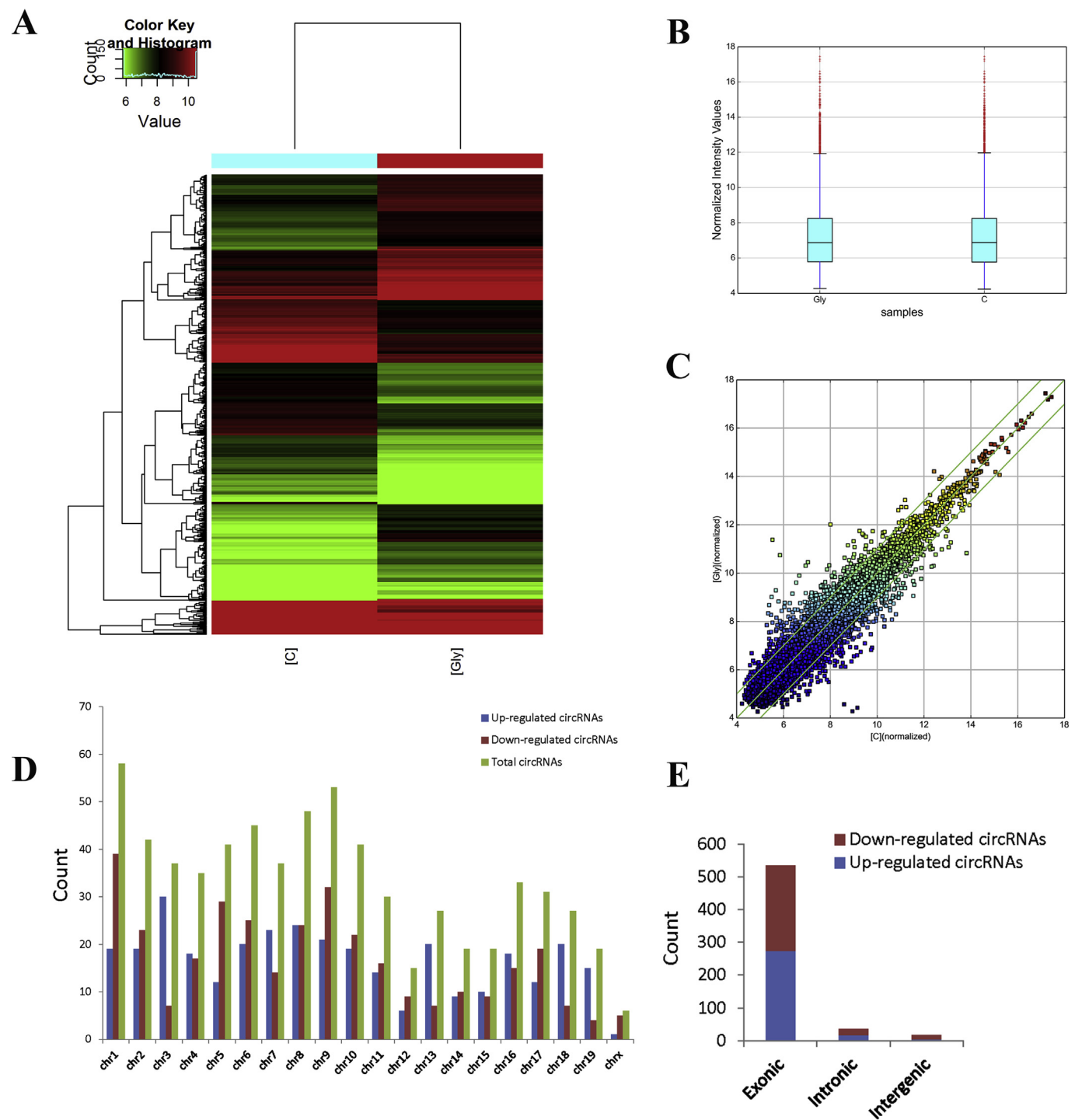
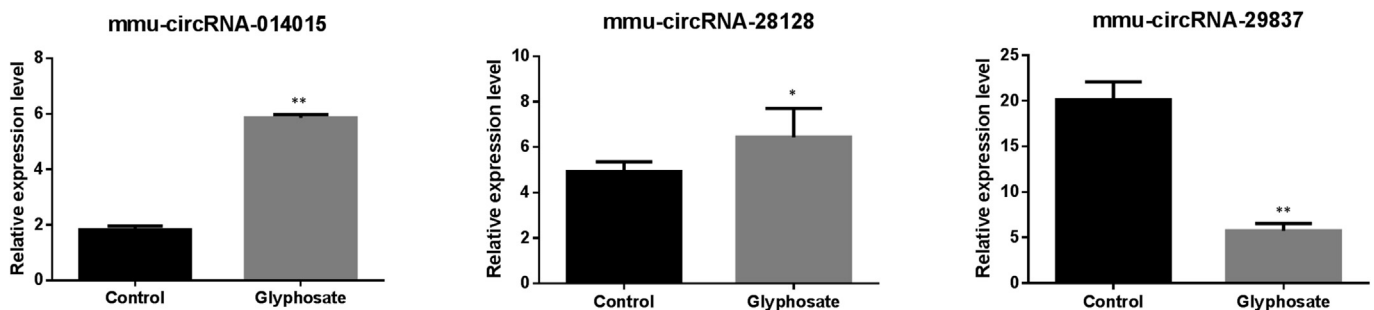


Fig. 1. Differential expression of circRNAs between perinatal glyphosate exposure samples and normal control samples by microarray. A: Hierarchical cluster analysis of all target circRNAs. We found that 663 circRNAs were differentially expressed ($aFC > 2.0$ and $p < 0.05$) between perinatal glyphosate exposure samples and normal control samples. Among these, 330 circRNAs were upregulated and 333 circRNAs were downregulated in perinatal glyphosate exposure samples. The color scale of the strips runs from green (low relative expression) through black (medium relative expression) to red (high relative expression). B: The circRNA profiles are shown as a box plot after normalization using the R software package. C: The raw variations in the circRNA expression profile between the two groups were assessed by a scatter plot. D: The chromosome distributions of differentially expressed circRNAs. E: Classification of differentially expressed circRNAs. C, normal control samples; Gly, perinatal glyphosate exposure samples. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 1

Top 22 upregulated and downregulated circRNAs between perinatal glyphosate exposure and control groups (signal value > 500, P-value <0.01 and aFC>3.6).

circRNA	Regulation	aFC	Chr	strand	type	GeneSymbol
mmu_circRNA_017140	up	16.09	chr16	+	exonic	Zbtb20
mmu_circRNA_35991	up	10.59	chr3	+	exonic	Vav3
mmu_circRNA_014582	up	10.3	chr3	+	exonic	Vav3
mmu_circRNA_43161	up	5.64	chr8	+	exonic	Inpp4b
mmu_circRNA_44739	up	5.29	chr9	–	exonic	Lca5
mmu_circRNA_32615	up	5.02	chr19	–	exonic	Slit1
mmu_circRNA_000340	up	4.9	chr9	–	exonic	Amotl1
mmu_circRNA_003071	up	4.81	chr19	–	exonic	Glis3
mmu_circRNA_23548	up	4.68	chr11	–	exonic	Dnah9
mmu_circRNA_014015	up	4.49	chr18	–	exonic	Arhgap12
mmu_circRNA_008286	up	4.14	chr13	–	exonic	Xrcc4
mmu_circRNA_007509	up	4.1	chr19	–	exonic	Slit1
mmu_circRNA_32318	up	3.98	chr19	+	exonic	Trpm3
mmu_circRNA_016710	up	3.98	chr6	+	exonic	Prdm5
mmu_circRNA_017868	up	3.82	chr19	+	exonic	Trpm3
mmu_circRNA_31221	up	3.79	chr18	–	exonic	Arhgap12
mmu_circRNA_26775	up	3.67	chr13	–	exonic	Iqgap2
mmu_circRNA_27713	up	3.64	chr14	–	exonic	Hmbox1
mmu_circRNA_41415	up	3.63	chr7	+	exonic	Nell1
mmu_circRNA_33633	up	3.62	chr2	+	exonic	Tanc1
mmu_circRNA_007990	up	3.61	chr7	+	exonic	Tead1
mmu_circRNA_28128	up	3.61	chr14	+	exonic	Itgbl1
mmu_circRNA_24151	down	7.94	chr11	–	exonic	Srcin1
mmu_circRNA_20127	down	7.57	chr1	–	exonic	Satb2
mmu_circRNA_20121	down	7.31	chr1	–	exonic	Satb2
mmu_circRNA_39855	down	5.9	chr6	+	exonic	Met
mmu_circRNA_29837	down	5.87	chr16	+	exonic	Abi3bp
mmu_circRNA_45659	down	5.82	chrX	+	exonic	Uprt
mmu_circRNA_012180	down	5.73	chr8	+	exonic	Phkb
mmu_circRNA_38710	down	5.67	chr5	+	exonic	Limch1
mmu_circRNA_37497	down	5.48	chr4	–	exonic	St3gal3
mmu_circRNA_000042	down	5.2	chr18	+	exonic	Prr16
mmu_circRNA_39509	down	4.77	chr5	+	exonic	Stx1a
mmu_circRNA_008614	down	4.32	chr9	–	exonic	Mlip
mmu_circRNA_30626	down	4.3	chr17	–	exonic	Satb1
mmu_circRNA_25343	down	4.27	chr12	–	exonic	Kcnh5
mmu_circRNA_45921	down	4.17	chrX	+	exonic	Phka2
mmu_circRNA_40615	down	4.15	chr6	–	exonic	Foxp1
mmu_circRNA_25345	down	4.04	chr12	–	exonic	Kcnh5
mmu_circRNA_018049	down	4.01	chr11	–	exonic	Cobl
mmu_circRNA_29625	down	3.93	chr16	–	exonic	Pdia5
mmu_circRNA_013750	down	3.87	chr1	–	exonic	Atp2b4

**Fig. 2. Quantitative real-time PCR validation of selected circRNAs.**

The relative expression of selected circRNAs (mmu-circRNA-014015, mmu-circRNA-28128 and mmu-circRNA-29837) in the perinatal glyphosate exposure group relative to the normal control group. *P-value < 0.05, **P-value < 0.01, vs control group.

3. Results

3.1. Expression profiling of circRNAs induced by perinatal glyphosate exposure in the hippocampus of mouse offspring

To describe the hippocampal circRNA profile in mice with perinatal glyphosate exposure, the microarray assay was performed to compare the circRNA expression between perinatal

glyphosate exposure samples and normal control samples. Hierarchical cluster analysis revealed the circRNA expression levels in the glyphosate exposure group and control group, and showed that these levels were different between the two groups. In total, 663 circRNAs were identified as differentially expressed (aFC > 2.0 and P value < 0.05). Among these, 330 circRNAs were upregulated and 333 circRNAs were downregulated in perinatal glyphosate exposure samples (Fig. 1A). The top 22 differentially expressed circRNAs

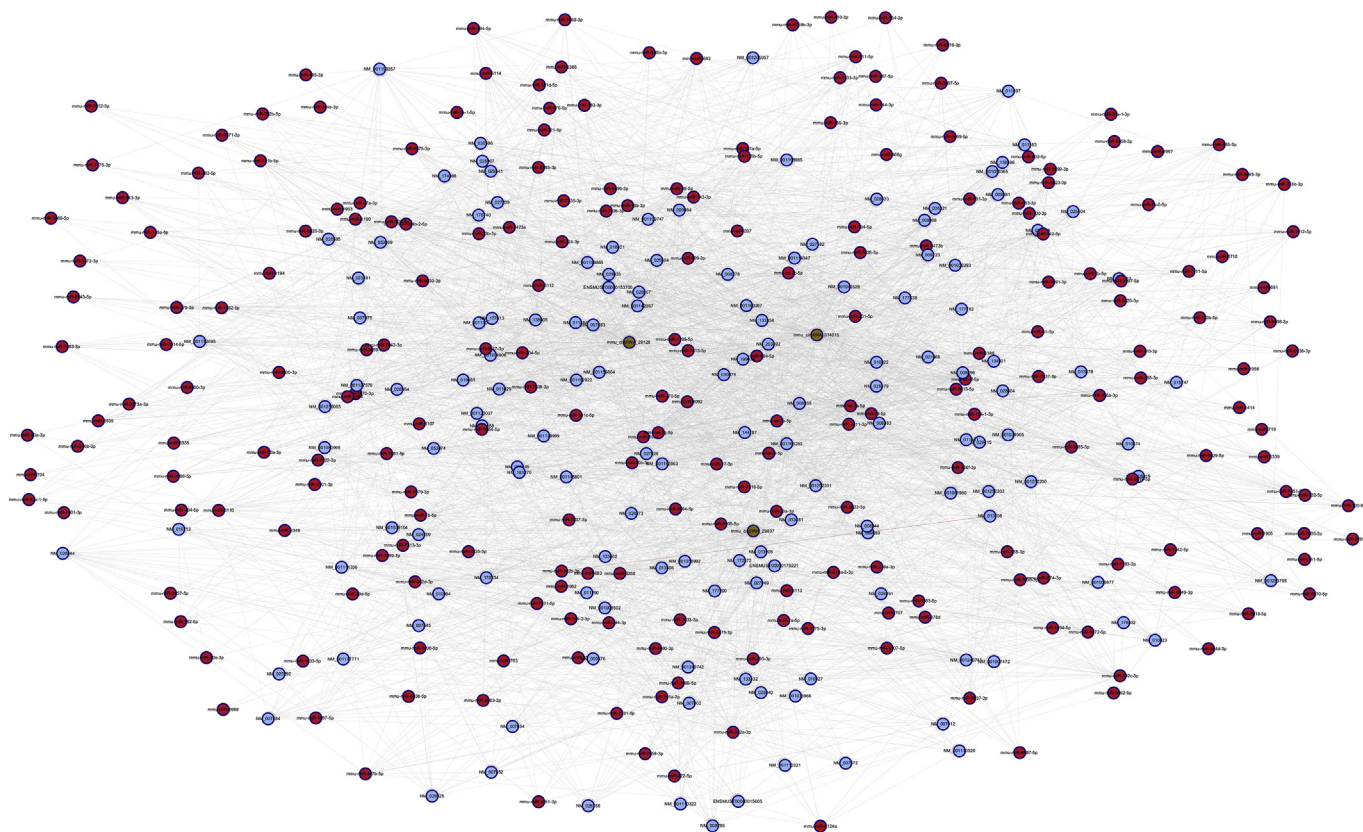


Fig. 3. CircRNA-miRNA gene network map from bioinformatics analysis.

There were 3 circRNAs identified that may play crucial roles in the hippocampus from mice with perinatal glyphosate exposure. Brown dots are circRNAs, red dots are miRNAs and blue dots are genes. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

are listed in Table 1 and Supplementary File 1. The circRNA profiles are shown as a box plot (Fig. 1B) highlighting the similar distributions for the two datasets. The raw variations in the circRNA expression profile between the two groups were assessed using a scatter plot (Fig. 1C). These data suggested that circRNA expression was different in the perinatal glyphosate exposure group and the control group. There was a wide distribution among the chromosomes (Chr) from which the differentially expressed circRNAs were transcribed, and the top 3 were Chr 1 (8.74%, 58/663), Chr 9 (7.99%, 53/663) and Chr 8 (7.24%, 48/663) in descending order (Fig. 1D). According to the genomic annotation, approximately 80.54% (534/663) of all differential circRNAs were exonic (Fig. 1E). The raw and analyzed microarray data, formatted as Microsoft Excel files, were deposited in the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO). The approved GEO accession number is GSE111670 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE111670>).]

3.2. Validation of candidate circRNAs using qRT-PCR

mmu-circRNA-014015, mmu-circRNA-28128 and mmu-circRNA-29837 were selected from the top 22 differentially expressed circRNAs and verified using qRT-PCR. The relative expression levels of mmu-circRNA-014015 and mmu-circRNA-28128 were upregulated by 3.22-fold and 1.31-fold, respectively, and mmu-circRNA-29837 was downregulated by 3.57-fold in the perinatal glyphosate exposure group compared to the control group. These results were consistent with the microarray results (Fig. 2).

3.3. Competing endogenous RNA (ceRNA) network analysis

The circRNAs, targeted miRNAs and mRNAs with expression levels that shared a meaningful correlation were subjected to ceRNA analysis. Then, a circRNA-miRNA gene network was generated to visualize the interactions using Cytoscape (Fig. 3). Annotations for the circRNA-miRNA interactions are depicted in Supplementary File 2.

3.4. GO and KEGG pathway analyses of the circRNA parental genes

To examine the biological functions of these differentially expressed circRNAs, GO analysis and KEGG pathways analysis were performed. In the GO analysis, the number of genes corresponding to GO entries was determined, and the enrichment score was reported as the $-\log(P\text{-value})$: For biological process, the term with the most genes was cellular nitrogen compound metabolic process (GO:0034641, count = 47), and the most significantly enriched term was organic substance metabolic process (GO:0071704, $P = 1.38E-04$) (Fig. 4A). For cellular component, the term with the most genes was cell part (GO:0044464, count = 90), and the most significantly enriched term was immunological synapse (GO:0001772, $P = 6.69E-04$) (Fig. 4B). Finally, for molecular function, the term with the most genes was binding (GO:0005488, count = 80), and the most significantly enriched term was tRNA methyltransferase activity (GO:0008175, $P = 3.40E-04$) (Fig. 4C). Similarly, the enriched pathways in the KEGG pathway analysis were aldosterone synthesis and secretion (mmu04925), ovarian steroidogenesis (mmu04913), steroid hormone biosynthesis (mmu00140) and relaxin signaling (mmu04926). These pathways

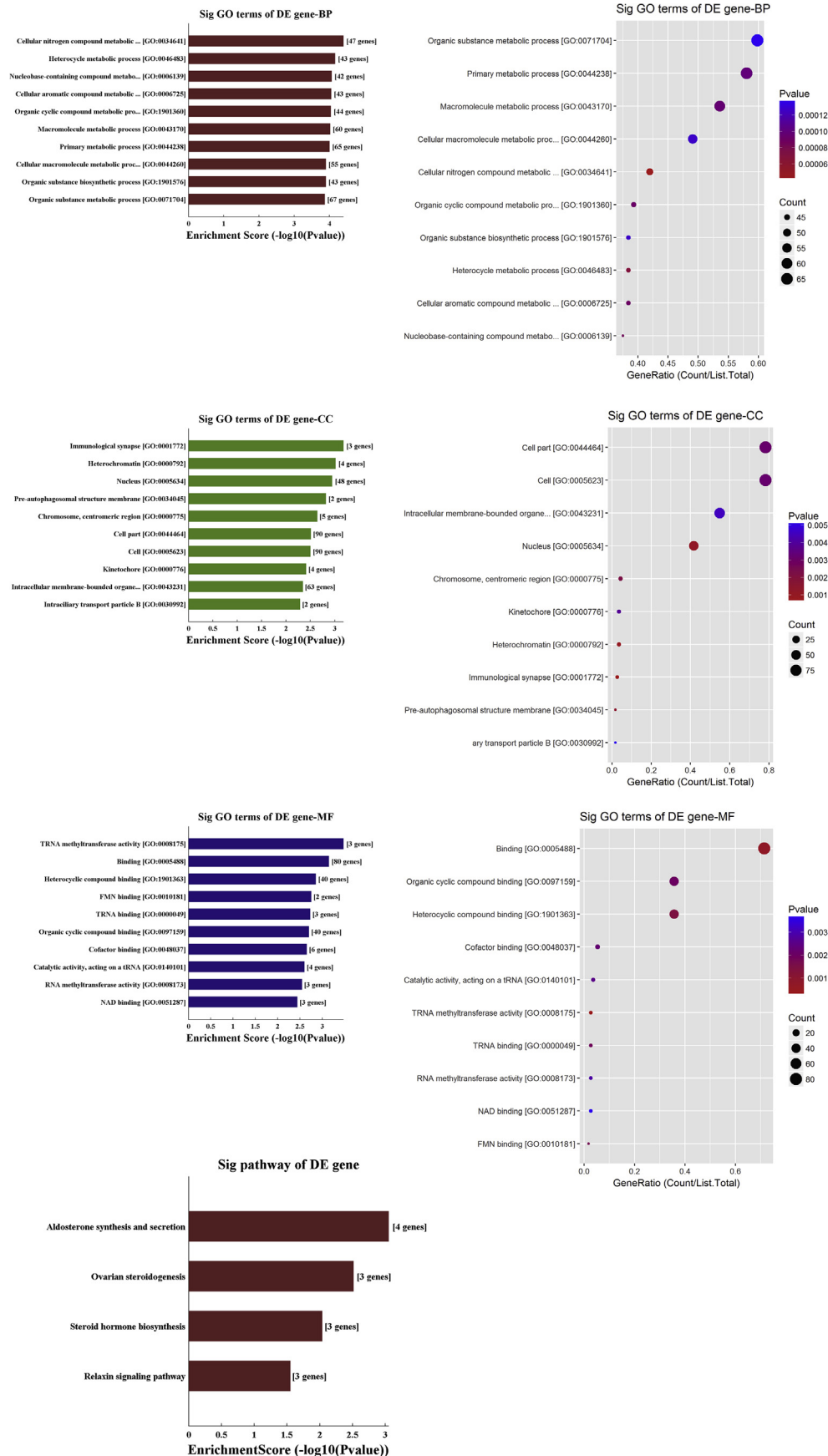


Fig. 4. GO and KEGG pathway analyses of mmu-circRNA-014015, mmu-circRNA-28128 and mmu-circRNA-29837 parental genes.

A: Top 10 GO terms of the gene count and enrichment analysis for biological processes. B: Top 10 GO terms of the gene count and enrichment analysis for cellular components. C: Top 10 GO terms of the gene count and enrichment analysis for molecular functions. D: The enriched KEGG pathways of the differentially expressed circRNA parental genes.

are related to stress-associated steroid metabolism and are most likely involved in glyphosate-induced neurotoxicity (Fig. 4D).

4. Discussion

Previous studies reported that exposure of adult rats to glyphosate could cause alterations in dopaminergic markers resulting in behavioral changes [1], while chronic glyphosate exposure of adult mice offspring during early development could lead to depression-like behaviors [17]. These behavioral alterations after glyphosate exposure may be associated with oxidative stress and glutamate excitotoxicity. Our previous studies indicated that mice with perinatal glyphosate exposure exhibited alterations of miRNAs and their target genes related to neurogenesis and neuron differentiation in the prefrontal cortex (PFC). We speculated that other factors may be involved, and these mechanisms are yet to be fully elucidated.

CircRNAs contain multiple miRNA binding sites and can serve as competing endogenous RNAs for miRNAs. Their interaction of circRNAs with disease-associated miRNAs indicates that circRNAs are important for the development and progression of nervous system diseases [18].

In this study, we investigated alterations in the expression of circRNAs in the hippocampus from mice with perinatal glyphosate exposure compared to normal controls. We found 663 circRNAs changed significantly, among them 330 were upregulated and 333 were downregulated. Furthermore, the differential expression of mmu-circRNA-014015, mmu-circRNA-28128 and mmu-circRNA-29837 was verified using qRT-PCR. This indicates that the expression of these circRNAs was affected by glyphosate. Thus, we propose that these three circRNAs may be involved in glyphosate-induced neurotoxicity. CircRNAs-miRNA gene network were identified by ceRNA bioinformatics analysis. We hypothesized that mmu-circRNA-29837 may function as a “sponge” for its targeted miRNAs in the circRNA-miRNA gene network prediction. The top 3 predicted target miRNAs are mmu-miR-7035-5p, mmu-miR-3083-5p and mmu-miR-34a-5p (Supplementary File 3). Previous studies have demonstrated that these miRNAs are involved in brain injury. For instance, miR-34a-5p could regulate neurite outgrowth and dendritic spine morphology [19], processes implicated in development of cognitive function [20] and neurodegenerative diseases [21]. A new study found miR-34a-5p can downregulate BDNF, which is involved in cognitive deficits, in the hippocampus through targeting its 3' UTR [22]. Our preliminary experiments indicated that the expression of BDNF and tropomyosin-related kinase receptor B (TrkB, a high affinity receptor for BDNF) in the PFC and hippocampus was downregulated by perinatal glyphosate exposure (data not shown).

We also used GO annotation and KEGG pathway analysis to functionally annotate the predicted target genes of mmu-circRNA-014015, mmu-circRNA-28128 and mmu-circRNA-29837. GO enrichment data revealed that the most significant GO terms were cellular nitrogen compound metabolic process, organic substance metabolic process, immunological synapse and tRNA methyltransferase activity. This indicates which coding genes contribute to glyphosate-induced neurotoxicity. In the KEGG pathway analysis, several important pathways closely associated with stress were identified, including aldosterone synthesis and secretion pathways. This suggests that these three circRNAs may participate in glyphosate-induced neurotoxicity through regulating stress-associated steroid metabolism pathways.

Previous research has found that subchronic glyphosate exposure of mouse offspring early in development could lead to depression-like behavior [17]. There are neuroendocrine changes in depression, including cortisol increase and HPA axis overactivity.

Aldosterone also plays an important role in the stress response and acts selectively in relevant mood-regulating brain areas [23]. These stress-associated hormones can control energy metabolism and stress-adaptation through the mineralocorticoid receptor (MR) and glucocorticoid receptor (GR) [24]. Our preliminary experiments also indicated that the expression of MR and GR in the PFC and hippocampus was downregulated by perinatal glyphosate exposure (data not shown).

In summary, this study identified that aberrantly expressed circRNAs are involved in glyphosate-induced neurotoxicity. It also showed that these circRNAs might affect expression of their target genes by regulating stress-associated steroid metabolism pathways. In the future, more investigations are needed to validate the interaction between circRNAs and their target miRNAs and genes. Finally, the roles of circRNAs in glyphosate-induced neurotoxicity will be investigated systematically according to the circRNA-miRNA gene network. This might help us to understand the mechanisms of glyphosate-induced neurotoxicity and find a new molecular target for clinical therapy or diagnostic tools.

Conflicts of interest

All authors declare that there are no conflicts of interest.

Author contributions

Design and conduct of the study: H. Ji.
Collection of the data: N. Yu, Y. Tong, D. Zhang, S. Zhao.
Management, analysis and interpretation of the data: H. Ji.
Manuscript preparation: H. Ji.
Critical revision and approval of the manuscript: All authors.

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Transparency document

Transparency document related to this article can be found online at <https://doi.org/10.1016/j.bbrc.2018.04.200>.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.bbrc.2018.04.200>.

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